



Creating an Emergency Radio System Using LoRa Technology

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Team Name: Yanny or LoRa

School: Eric Hamber Secondary School

City: Vancouver, Canada



ABSTRACT

The advanced CanSat Design Challenge is an annual spring competition where Canadian high school students design a can-sized satellite, which is then launched 1 kilometre by a rocket in Lethbridge, Alberta. Our team, *Yanny or LoRa*, is composed of 6 high school students from a high school in Vancouver, Canada. Using 3D technology, we constructed a satellite that fulfills two main missions. Our standard primary mission is to collect air pressure, humidity, and temperature data, and to transmit this data through radio back to the ground station. Our secondary mission is to use LoRa (Long Range) technology to create an emergency contact radio system. Our emergency device sends information (specifically SOS signals, heart beat, and GPS location) to the CanSat, which then relays the information back to the ground station. The CanSat allows the message to send through long distances, which we hypothesize to be several kilometres. This store-and-forward system mirrors the communication techniques used in real life satellite systems and is a practical application of aerial LoRa nodes as temporary communication in remote or disaster-affected areas. As the CanSat collects this data, it descends back down using a parachute.



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1. INTRODUCTION

1.1. Team Mission and Objectives

Our primary mission is to collect air pressure, humidity, and temperature data using a BME280 sensor. This information is collected every second and transmitted through radio to the ground station.

Our secondary mission demonstrates a store-and-forward emergency communication relay system using LoRa radio technology. The concept is that a person in a remote or dangerous location could use a low-power LoRa device to transmit an emergency message to the CanSat, which then relays that message to a home base receiver kilometres away. This has extremely practical effects for a person in a location where they cannot access traditional signal or radio infrastructure. The LoRa technology used for the CanSat extends the effective range of the ground station radio beyond what would otherwise be possible.

Many areas, including Alberta (where we launch our CanSat), have long wildfire seasons that may destroy ground infrastructure and cellular towers. Our CanSat explores the real world problem of finding the minimum viable aerial relay node for emergency communication. Contrary to existing satellites (such as Starlink), our project aims to provide long-range communication *without* relying on existing infrastructure or



large-scale setups. Our aim is to create a useful SOS system that anyone can deploy and use at low cost.

1.2. Team Members and Roles

Our team consists of six high school students from Eric Hamber Secondary School, which is a public school in Vancouver, Canada.

Asa Liu (co-captain)

Asa Liu is a grade 11 student at Eric Hamber Secondary who plans on pursuing a future in becoming an astronaut. He has an incredible drive towards building new projects centered around controlled movement. Asa's background includes many Arduino-based projects and participating in physics competitions. His role in the team is planning and organizing the CanSat build. He is also leading the secondary mission (LoRa development) and designing the frame + hardware systems.

Matthew Song (co-captain)

Matthew Song is a grade 11 student who plans to pursue aerospace or mechanical engineering at university. He has a strong interest in engineering and safety design, and is currently working on an engineering project centered around improving the protective performance of football helmets. Besides academics, he also enjoys golfing and watching football. On the team, Matthew is responsible for developing and testing the landing gear. He will also run meetings and make sure that everyone is on pace to finish.

Zoey Cheang

Zoey Cheang is a grade 12 student who plans on pursuing computer science in university and work in the cybersecurity industry in the future. She has participated in multiple Capture The Flag competitions and is working toward building a strong foundation in networking protocols. She is comfortable coding with Python, C++ and Java. Zoey will be focusing on working on the three-way communication LoRa protocol, electronics hardware, and creating a website for the team.

Naiming Zheng

Naiming Zheng is a grade 11 student at Eric Hamber Secondary School who plans to pursue computer science at university. He is proficient in Python and Java, and he is currently trying to learn Rust to get a deeper understanding of system programming and memory safety. Naiming is also in Eric Hamber's cybersecurity club and competes in Capture the Flag Competitions. Naiming is responsible for the software side of the primary mission, focusing on sensor data processing and controller logic.



Dora Yuan

Dora Yuan is a grade 11 student at Eric Hamber who is passionate about engineering, and is planning on pursuing it in university. She has a strong background in physics, is proficient in Python, and is currently expanding her knowledge on electronics and CAD. On the team, she will be leading the outreach, as well as working on the CanSat landing gear. She is the main writer of this report.

Bogdan Shkromiuk

Bogdan Shkromiuk is a grade 12 student at Eric Hamber who plans to pursue electronic engineering at university. He has a strong physics background and is experienced with 3D modeling, printing, and soldering. He will use these skills to build the CanSat frame and help with electronics.

TEACHER: *Mark Lam*

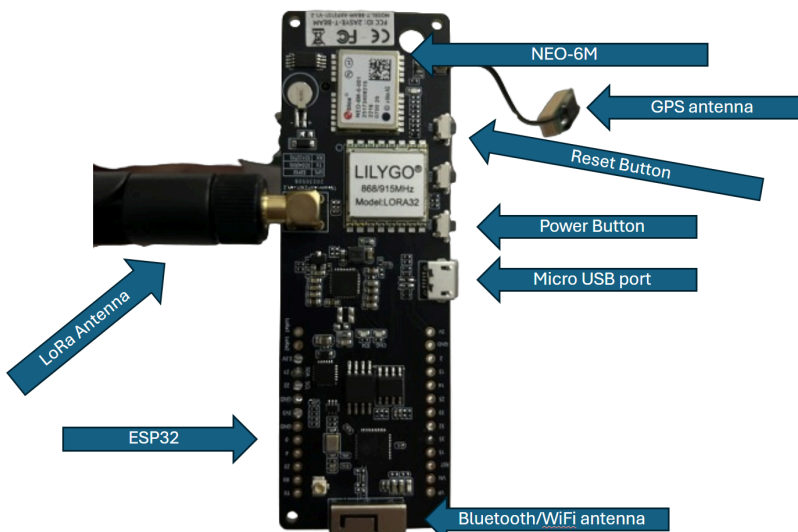
Mark Lam is this team's teacher and an Eric Hamber alumnus who graduated from UBC with a degree in physics. At Eric Hamber Secondary, he teaches many physics courses, including both AP Physics C classes. He has been doing CanSat with his students for many years, on top of supporting his class in various other physics events and opportunities.

2. CANSAT DESCRIPTION

2.1. Mission Overview (Main Components)

LilyGO T-Beam V1.2

Figure 1: LilyGO T-Beam





There are three main devices in our missions, CanSat, ground station, and SOS radio device. Our central component for all three devices is a LilyGO T-Beam V1.2. This contains a microcontroller, a LoRa module, a GPS module, an antenna, and a built-in battery connector.

Microcontroller

The microcontroller we are using is an ESP32. The board is mounted in the middle platform of the CanSat structure using plastic layers and screws. We have also used standoffs to prevent electrical contact between circuit board and base to prevent electrical short circuits.

LoRa Module

The LoRa module we are using is SX1276. It is designed specifically for long range, low-powered wireless communication, which greatly suits our missions. In our missions, it will be operated in the 915MHz ISM band.

GPS module

We are using the u-box NEO-6M GPS module. It provides geographic coordinates and altitude, helping us to track the CanSat location.

Antenna

We are using a 915 MHz LoRa antenna to match the frequency we are using for our LoRa transmitter. The antenna is mounted with the SMA male of the T-Beam module.

Battery

We will be using the LiPo 18650 flat-top cell 3.7V battery. It provides enough voltage to power the ESP32 and other sensors. The battery is housed in the integrated holder on the underside of the T-Beam. It is rechargeable via the micro-USB port integrated directly onto the T-Beam board.

CanSat

The main parts of our CanSat consists of a LilyGO T-Beam V1.2 mentioned above, a LED, a sensor, a power switch, and a parachute.

Figure 2: Our CanSat



Sensor

We are using a BME280 sensor, which tracks temperature, air pressure, and humidity. The sensors will be mounted on the same platform as the microcontroller. There will be holes to allow for air to flow in without endangering the sensor.

LED

The LED is used as an indicator when the CanSat is powered on. It is clearly visible on the side of our CanSat.

Power Switch

The power switch is clearly visible on the side of our CanSat, alongside with the LED to power on the CanSat.

Parachute

Our circular parachute system is attached to the top of the CanSat structure. The attachment point is reinforced to withstand the forces experienced when the parachute deploys. The parachute lines connect to the top cap so that the CanSat remains upright during descent, ensuring the antenna maintains good communication with the ground station. More information about the parachute can be found below, in 2.5. Recovery System.

SOS radio device



The main components of our SOS radio device consists of a BME280, a heartbeat sensor, a buzzer, a power switch, a LilyGO T-Beam V1.2 (mentioned above), SOS button, and an OLED screen.

Figure 3: Radio device



OLED Screen

The OLED screen displays BME280 data (Altitude, Humidity, Pressure, Temperature), GPS location, and heartbeat (in BPM). When an SOS message is sent, the OLED screen will indicate to the user that the message is being sent. When the SOS radio device receives an ACK message from the CanSat, the OLED screen will display to the user that the message is sent successfully.

BME280

The BME280 sensor allows us to collect temperature, humidity, altitude, and air pressure allowing the ground station to see the surrounding conditions of the SOS Radio.

Heartbeat Sensor

Using a simplistic 3d frame design we are able to collect the heart rate (bpm) of any user that puts their finger on top of the carefully designed frame that rests on top of the sensor. When measuring your heart rate the OLED screen automatically switches to heart reading mode and displays your bpm in big visible numbers.

LILYGO T-Beam v1.2

The LoRa32 microcontroller is the core of the SOS device. It is connected to all the parts and processes both sensors while communicating with the CanSat.

Buzzer



The buzzer beats to the BPM and is only active when a finger is on the sensor.

Power Switch

Turns on/off the SOS Radio.

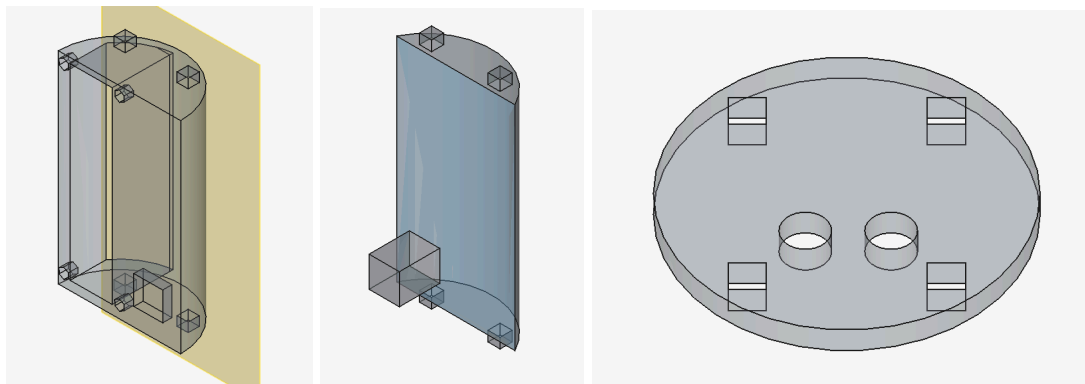
SOS Button

Sends an SOS signal to the CanSat when double-clicked. Double-clicking again cancels the SOS. Sends the GPS location and the heartbeat of the radio device user.

2.2. Structural Design (Frame)

For the final frame design, we went with a 4-part frame with 2 body parts along with a top lid and bottom lid.

Figure 4: Frame



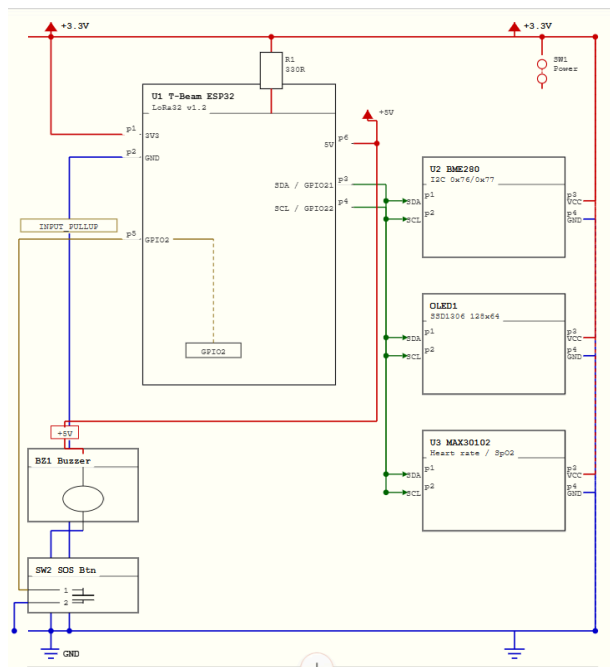
The diameter of the frame is exactly 66mm, and the height of the CanSat is 115mm (105mm body and 5mm lids). The interior design had to fit the 65mm LoRa32 + antenna, so we chose not to add walls on the side but instead reinforce it using the top and bottom. This also allowed for easy placement of switch, LED, and antenna holes.

The top lid includes a hole for the GPS to reach open sky while the bottom lid includes a circular hole for air to reach the BME280. Inside the body, we implemented 4 screw holes to tightly secure the LoRa32 to the frame.

Finally two holes were implemented on the top for parachute connection. We made sure the top lid was tightly sealed on the CanSat body so that it wouldn't fly off during parachute activation.

2.3. Electrical Design

Figure 5: SOS radio device wiring



BME 280

- Uses I2C (SDA=GPIO21, SCL=GPIO22)
- Connected to 3.3V and GND

OLED screen

- Uses I2C (SDA=GPIO21, SCL=GPIO22)
- Connected to 3.3V and GND

MAX30102

- Uses I2C (SDA=GPIO21, SCL=GPIO22)
- Connected to 3.3V and GND

Buzzer

- Connected to 5V and GND

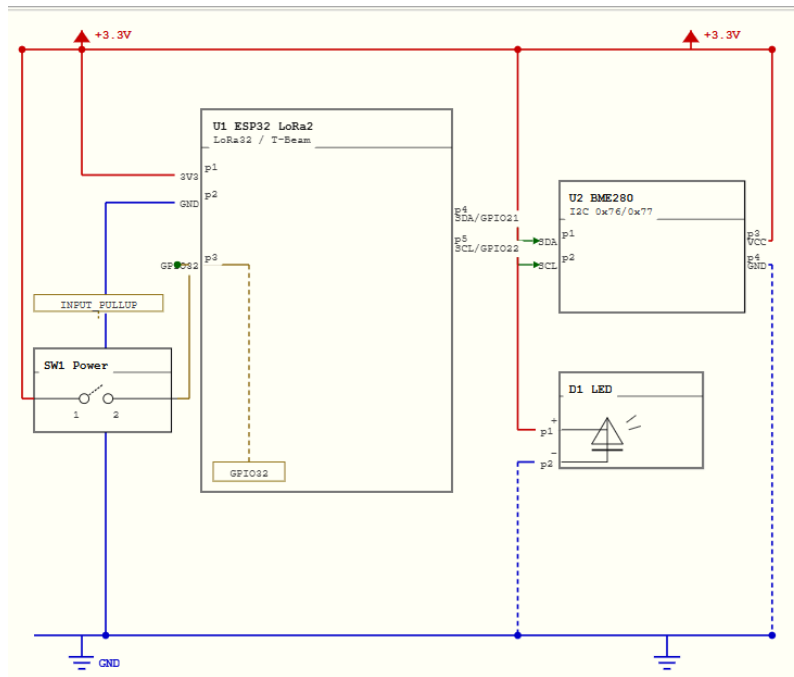
SOS button

- Connected to GPIO2 (INPUT_PULLUP) and GND

Toggle switch

- Connected in series with BME280 and 3.3V

Figure 6: CanSat device wiring



BME280

- Uses I2C (SDA=GPIO21, SCL=GPIO22)
- Connected to 3.3V and GND

LED

- Connected from 3.3V to GND so it turns on while the LoRa32 is on

Toggle switch

- Connected between GND and GPIO32

Input Pullup

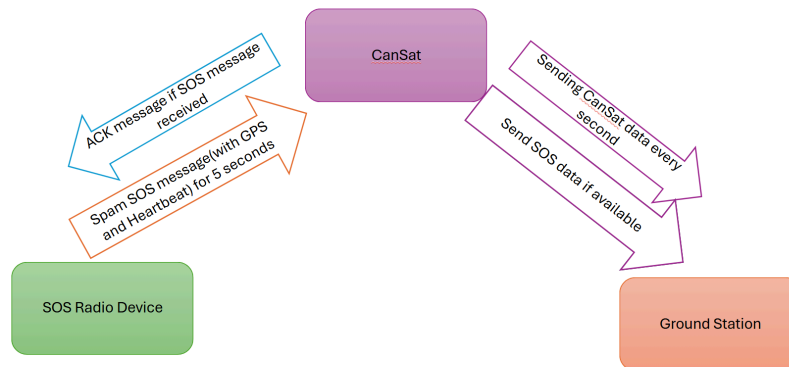
- Acts as an internal resistor to bring down voltage to 3.3V

Battery

- Powers everything

2.4. Software Design

Figure 7: Communication Logic

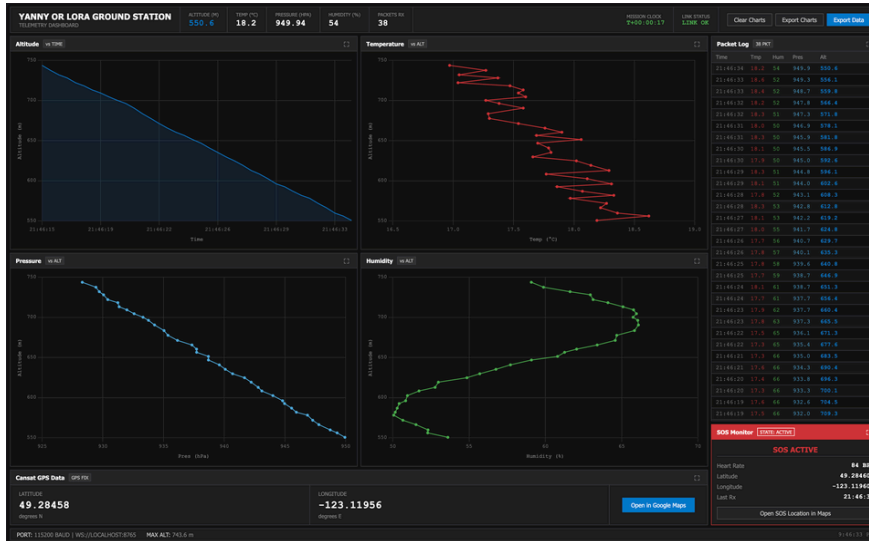


Our secondary mission uses three-way communication to communicate between the SOS radio device, CanSat, and the ground station. Since LoRa is not able to receive (RX) and send data (TX) at the same time, the CanSat would have to alternate between RX and TX mode to communicate with both the ground station and the SOS radio device. With the primary mission requirement, we are required to send CanSat data to the ground station every second. In another word, the CanSat will have to alternate between RX and TX mode within one second. To ensure the SOS message is successfully received by the CanSat, we use the following logic:

1. The SOS radio device spams an SOS message to the CanSat for 5 seconds, then immediately changes to RX mode.
2. The CanSat is in RX mode for around 0–850ms.
3. The CanSat is in TX mode at around 850–910ms. (Relaying SOS message if available, and sending CanSat data to the ground station)
4. The CanSat sends an ACK message (acknowledgment message) to the SOS radio device after successfully receiving the message.
5. The CanSat is in TX mode (ACK if available) at around 910–925ms.
6. The SOS OLED screen displays that the message is successfully received by the CanSat.
7. If the SOS radio device does not receive an ACK message from CanSat, the SOS message is spammed again for 5 seconds to the CanSat.

Using the logic above, the CanSat is in RX mode 80% of the time, ensuring the SOS data is successfully received and relaying back to the ground station.

Figure 8: Dashboard



Our dashboard, which is at the ground station, features several information:

1. An altitude vs time graph
2. An altitude vs temperature graph
3. An altitude vs air pressure graph
4. An altitude vs humidity graph
5. The GPS, temperature, pressure, and humidity data of the CanSat
6. The GPS coordinates of the emergency device (in longitudes and latitudes)
7. An SOS message monitor, which alerts when an SOS signal is received

Our dashboard makes it so that it is easy to search up the GPS coordinates on Google Maps. We originally wanted to include a static map within the dashboard itself, but we couldn't obtain a license to download one in time. For future prototypes, we hope to be able to include a map. The dashboard is created in HTML and using python and javascript, we are able to connect to the ground station serial monitor to display the data in our dashboard.

2.5. Recovery System

Our recovery system is a parachute to slow down the CanSat's descent. Our parachute is made of nylon fabric. It has a diameter of 0.4 m, but there is a hole in the middle with a diameter of 0.12 m, so the area of the parachute is 0.114m². After the CanSat is launched, the parachute allows it to generate drag force as it descends back down. Drag force is described by the following:

$$F_D = \frac{1}{2} \rho v^2 C_d A,$$



Where ρ is air density, v is descent velocity, C_d is the drag coefficient, and A is the area of the parachute. At terminal velocity, drag force is equal to the gravitational force on the CanSat (mg). According to our research, circular parachutes have a drag coefficient of approximately 1.75, but since there is a hole in the middle of ours, we assume that our drag coefficient will be 0.8. The density of air at sea level is approximately $1.229 \frac{kg}{m^3}$.

At terminal velocity, the net force is 0. Since the two forces acting on the parachute are drag force and gravity, the terminal velocity of the parachute can be calculated:

$$\begin{aligned} F_D &= F_g \\ \frac{1}{2} \rho v^2 C_d A &= mg \\ v_T &= \sqrt{\frac{2mg}{\rho C_d A}}, \end{aligned}$$

Where v_T is the terminal velocity in $\frac{m}{s}$, g is $9.8 \frac{m}{s^2}$, and m is the mass of the CanSat, which is 0.301 kg. By plugging in our numbers:

$$v_T = \sqrt{\frac{2(0.301)(9.8)}{(1.229)(0.8)(0.114)}} = 7.24 \frac{m}{s}.$$

Our approximate terminal velocity $7.24 \frac{m}{s}$, which is within the required descent velocity.

Our initial design for the parachute was made using plastic garbage bags, which we chose for its light weight and ability to deploy reliably. However, during testing, the material was found to be too fragile, as the parachute tore upon impact with a tree. In this initial configuration, the parachute had a diameter of 140 cm and produced an estimated descent rate of 2 m/s, which was well under the recommended minimum velocity. Consequently, a second design was developed using nylon fiber, which was strong enough to not rip during the launch or descent. For the revised parachute, the diameter was reduced to 40 cm, resulting in an increased descent rate of approximately 4 m/s, which significantly reduced the horizontal drift. However, upon arrival in Alberta, we realized that the wind speeds were approximately 50 km/h, which would drift the CanSat more than 5 km away from the launch point. To address this issue, a 12 cm diameter vent was added to the centre of the parachute. The estimated increase in descent rate would keep the parachute within a reasonable distance from the launch site, while still keeping it within the maximum recommended speed from the rulebook.

The string that attaches to the CanSat is evenly spread out across the edge of the parachute, connected at 8 points to make sure the parachute is symmetrical. The string is



made of polypropylene twine; one of the requirements is that it had to withstand 30 Newtons of force, and we found this to be the strongest option. Unlike the braided fishing line, which we also considered, the polypropylene twine did not tangle as easily.

Figure 9: Parachute



3. PROJECT PLANNING

3.1. Budget

Table 1: Budget

Component	Quantity	Cost per Item (CAD)	Total Cost (CAD)
LILYGO T-Beam V1.2 ESP32 LoRa Board	3	\$49.98	\$149.93
BME280 Temperature & Pressure Sensor	1	\$19.99	\$19.99
HiLetgo MAX30102	1	\$12.99	\$12.99
915 MHz LoRa Antenna 5dBi DIYmalls	3	\$6.99	\$20.98
LiPo 18560 flat cell battery	3	\$23.99	\$71.97



Component	Quantity	Cost per Item (CAD)	Total Cost (CAD)
Heat Shrink	1	\$3.60	\$3.60
Mini Toggle Switch	1	\$1.65	\$1.65
PLA Filament (3D Printed Frame)	1 kg	-	\$25.99
Parachute Materials	2	\$16.49	\$32.98

Total Cost: \$340.08 CAD

3.2. Testing Process

Our testing process took the majority of the time, and started after we got our parts.

The first test was the hardware testing where we tested the BME280 sensor, heartbeat clicker, and GPS module. For some members of our group, it was the first time using Arduino and soldering, so we all took time learning it. To test them, we first had to solder header pins to the LoRa32 board.

After, we tested the communication for the LoRa and walked in different directions to test its range. After confirming a range of around 500–700 metres in dense urban regions with hills, we transferred the BME280 data between the two LoRa32s.

The third step was to fully wire the SOS system and the CanSat system to see how each part interacted with each other. At this time, we also began designing the frame and parachutes.

When we finished breadboarding and tested each device individually and in pairs, we split into 4 teams. One team would solder the wires together, one team would work on the parachute, one team would 3D design the frame, and the last team to finish the 3-way communication code.

After the parachute was finished, we began using our failed frames with weights to test its falling velocity. We walked up a hill and dropped the parachute from 10 meters high. We divided the distance, which was 10 meters, by the average time taken for descent to find the descent velocity. To ensure that the parachute opened under any circumstance we threw it in many different configurations at a variety of linear and rotational speeds.



Once the frame was finalized, we tested its energy dissipation capabilities by dropping it against the floor at different heights and measuring the total energy before and after its collision with the ground. These tests ensured that we had a frame that best protected the components inside during both the launch and the impact with the ground after descent. This was necessary considering the potential 20 g's of force the CanSat may endure.

When everything was soldered, we could finally test the 3-way communication. This proved to be incredibly hard and took a long time to get working, because the CanSat could only either receive or send signals, not at the same time. During the testing, we also started placing components in the SOS radio frame and CanSat frame along with the parachute.

Finally, we finished all the testing and sealed the lid one last time. We learned that instead of using a breadboard and headerpins, we should have used a custom PCB and solder directly because it would have saved hours of unsoldering and rewiring. We originally had a separate battery, but the LoRa32 came with a built-in battery holder and we had difficulties unsoldering the original. This made creating the frame incredibly difficult, because we had to keep remaking it.

4. OUTREACH PROGRAM

Our outreach program consisted of both social media and in person presentations.

For social media, we made a website to explain our project, as well as an Instagram page that regularly updated with stories and posts to illustrate our process. Our Instagram has several hundred followers.

Table 2: Summary of social media activities

Platform	Link	Description
Website	https://yannyorlora.tech	CanSat information + mission homepage
Instagram	https://www.instagram.com/yannyorlora	Work-in-progress updates (on our story), informative posts, record of in-person outreach activities
LinkTree	https://linktr.ee/yannyorlora	Central hub for social media information



For our in person outreach, we gave three presentations to various audiences. The first presentation was done to a high school audience, in a STEM-focused school club called Experiment Club. This audience (grade 8–12) was very fitting because participants were already interested in science and engineering. Our presentation focused on explaining what CanSat is, as well as running an activity on circuits and breadboards. We taught students how to use breadboards to create several types of circuits, using LEDs. We then explained how we use solderless breadboards to help us test the CanSat in its design stage. Several of our participants expressed interest in the CanSat competition.

Our second in person presentation was given to an elementary school audience, to a grade 4 class in a public elementary school called Quilchena Elementary. We talked about our CanSat to the class, and then ran an egg drop activity. During our time there we spoke with some individuals with incredible talent with the same catchment as our school so we hope to see them participate in CanSat in the future.

Our third in person presentation was given to a public audience at our local community centre, called Kerrisdale Community Centre. We gave our presentation to a class of kids aged 9–12 in a STEM program (this is a public program that allows both registration and drop-in). We presented our CanSat, and then ran an egg drop activity.

Figure 10: Pictures from our in person outreach events



5. APPENDIX: FRAME DESIGN PROCESS

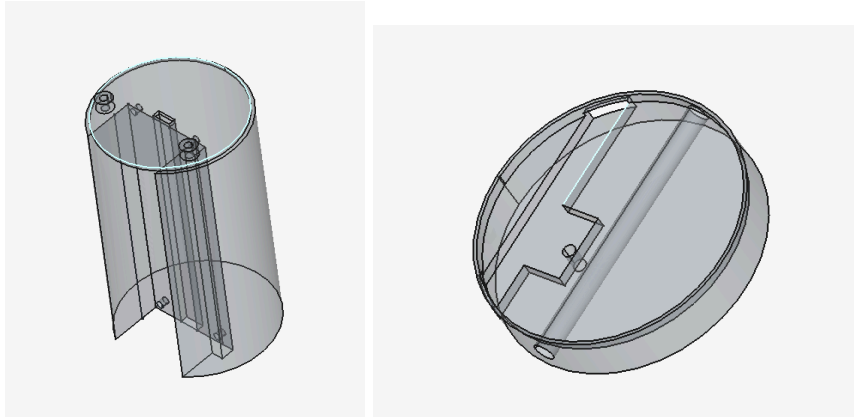
The goal of our frame design was to develop a casing that fit the 66mm x 115mm size while leaving room for the LoRa32 and antenna. It also had to fit all the parts in a secure way so that on the launch up and during the descent the parts wouldn't be moving around. The



launch force was estimated to be 20gs and the landing speed was estimated to be around 7m/s. This was our basis before designing the frame.

The material we used during printing was PLA, and the software used for the design is FreeCAD. Below are some drafts we made during the design process.

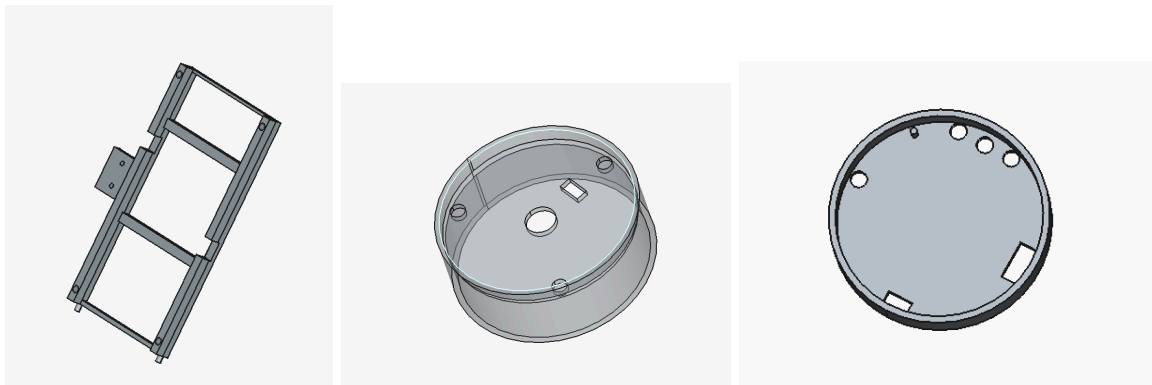
Draft 1 (Unfinished)



The idea was to have half the circle be 3D printed so that we could change the weight to counter the battery and LoRa32 weight. It had a cut-out hole and screw holes to secure the microcontroller. This was our first time 3D designing.

Missing: proper dimensions, LED hole, charger hole, battery hole, switch hole, parachute holder.

Draft 2 (unfinished)



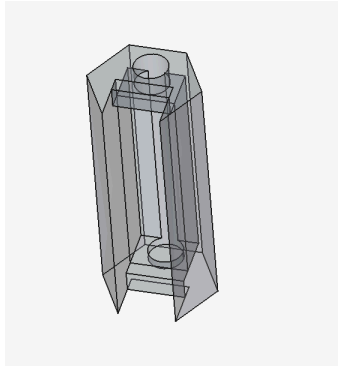
This was completely different from the original design. We printed a standalone middle frame to act as a slide holder for the LoRa32. This design left space for wiring and components to be on the side. The middle standalone frame also included screw holes for



the CanSat and a side platform for the BME280 sensor. There were 3 holes on the bottom frame to allow air to reach the sensor.

Missing: proper battery holder, side walls to protect the centre, incorrect charger placement.

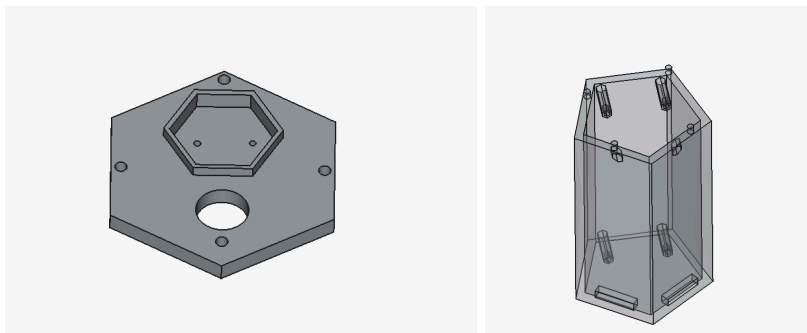
Draft 3 (unfinished)



For this design we tried something new: a single piece design with a sliding door for battery access and an antenna channel through the centre of the body along with mounting bars for the top and bottom. For this design, the BME280 sensor was planned to be taped to the interior side of the wall.

Missing: charging hole, parachute holder

Draft 4

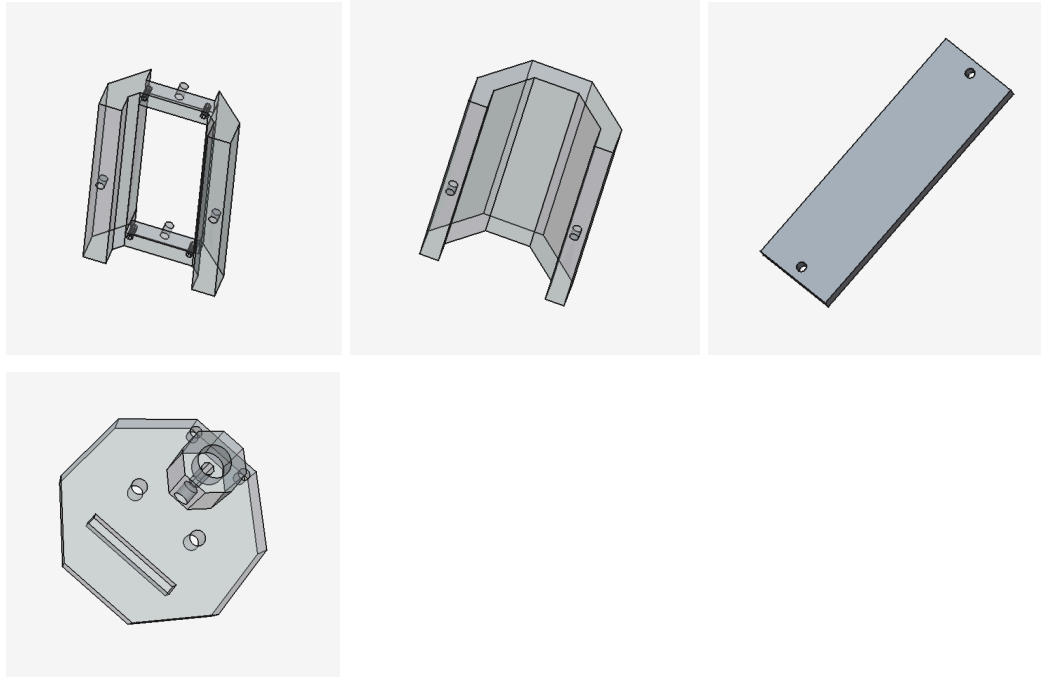


This was the most complete design so far. Changes included a pentagon body shape which allowed for flat faces on the sides making it easier to mount hardware and add features compared to a circular body. There are 4 pegs to mount the LoRa32 while leaving space behind for battery and wiring. Air vents were also added to let in air for more precise readings. The top lid was designed to have an antenna hole and a parachute holder with 3 secure holes to allow for parachute knotting.



Missing: BME280 mounting area, charging hole, battery switch hole, antenna mount

Draft 5



This was the first complete design with all the proper parts to secure the components. This had a 4 part system where the two body parts would snap together before securing the top lid and bottom lid (no image) sandwiching the body. This was a massive improvement as it left space for battery change and had a dedicated antenna holder and a case to close the battery. After printing, many flaws were found, such as wrong sizes for the antenna hole and battery holder. Securing pegs were also too small and often snapped.

Missing: proper measurements.

Version 6 (Final Version)

More information about this frame can be found in 2.2. *Structural Design (Frame)*.

During our testing, we discovered that the antenna had to be connected to the SX1276 dedicated SMA port on the board, and we did not have an extension. This meant that the board and antenna would be exactly 65mm, which was a problem. Instead, we ditched the side walls and focused the strength on the top and bottom.



In our design, we have a place to screw the LoRa onto the body with an open battery holder to hold and change the battery without taking it apart. The right side of the body was left for the antenna, and the extra room on the bottom left was used as extra security. The top and bottom frame were nearly identical except for a small hole for the GPS module.

This design worked well, but the wiring we used was too big and it took a while to fit everything together. The best part about this design was that it allowed lots of room on the side for the switch, LED, charger port, and sensor.

The main lessons we learned from this design process are the following:

1. Tolerance: had to be at least .25mm and .200 for perfect fit PLA.
2. Pre-planning: If we were to design it again, planning out where all the components should be placed first would allow us to be more efficient. PCB designing would have also saved us a lot more space rather than tons of wiring.
3. Functionality: For the second and third design, we tried to make it look more aesthetic, but it was not actually practical. We learned functionality should be prioritized.